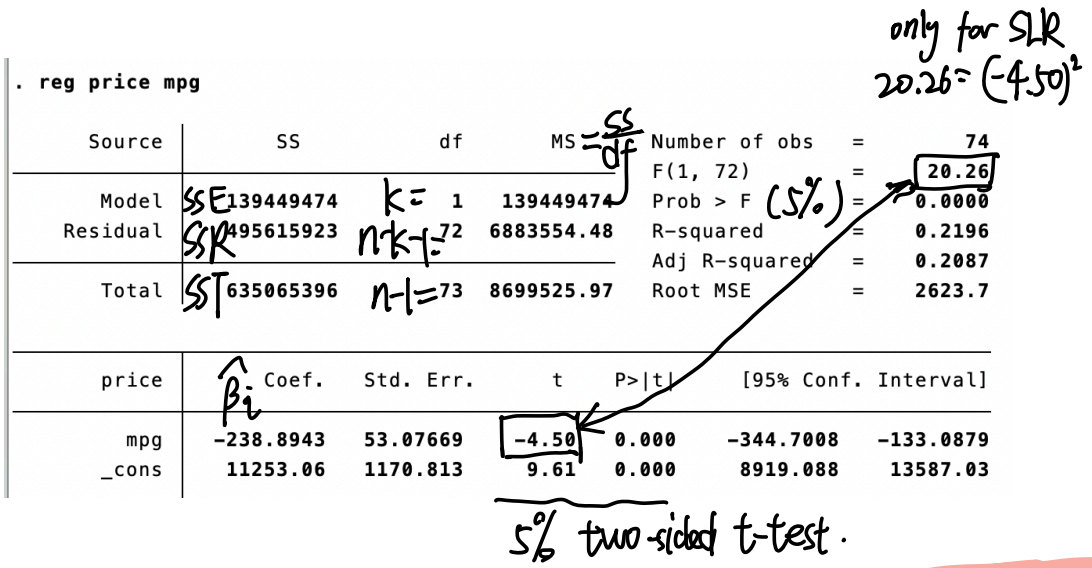
ECO3121 期中复习资料（2022版，基于对往届资料的整理补充而来）

1.考试相关信息（从林炜老师处获得，请以自己老师提供的信息为准）：考试范围包括所有已学内容；题型为问答题，无选填；不考STATA 操作，但会考察对STATA输出的表格（如图）的理解；不会有复杂计算，涉及到的计算笔算即可。
2.编写的同学水平有限，难免有所纰漏，请以老师所讲内容为准。
3.若笔记中出现课上未涉及内容，请忽略。



```
. reg price mpg

      Source |       SS           df       MS      Number of obs   =        74
-------------+----------------------------------   F(1, 72)        =     20.26
       Model |  139449474         1  139449474     Prob > F        =    0.0000
    Residual |  495615923        72  6883554.48     R-squared       =    0.2196
-------------+----------------------------------   Adj R-squared   =    0.2087
       Total |  635065396        73  8699525.97     Root MSE        =    2623.7

------------------------------------------------------------------------------
       price |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         mpg |  -238.8943   53.07669    -4.50   0.000    -344.7008   -133.0879
       _cons |   11253.06   1170.813     9.61   0.000     8919.088    13587.03
------------------------------------------------------------------------------
```


F-test: $H_0: \beta_1 = \beta_2 = \dots = \beta_i = 0$. Because this is a SLR model.

$H_0: \beta_1 = 0$, $F_{1,n-k-1} = t^2_{n-k-1}$

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

$$\text{adjusted } R^2 = 1 - \frac{SSR/(n-k-1)}{SST/(n-1)} = 1 - \frac{\hat{\sigma}^2(n-1)}{SST} = 1 - \frac{(1-R^2)(n-1)}{(n-k-1)}$$

The adjusted R-squared imposes a penalty for adding additional independent variables to a model.

Important Notes on Midterm (from Prof. Lin Wei)

- Closed book. Largely test your intuition and interpretations.
- Time: October 30th, Sunday evening 7:00-9:00 PM.
- Location: TBD (will be posted via blackboard)
- If you have physical education test, No worries!
- You can come no later than 8 pm and still have two hours to finish the exam.
- Earliest submission: no earlier than 8 PM
- A cheating sheet containing all the necessary (more than) equations and formulas will be provided.

No need to prepare CS by yourself

Ch 1.

1. When writing a econometric model. don't forget "u"

   eg: $wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 training + u$

2. elements. variables and observations.

3. economic data sets
   - Cross-sectional data (横截面数据)
   - Time-series data (时间序列数据)
   - Pooled cross sections (混合横截面数据)
   - Panel/Longitudinal data (面板或纵列数据)

（2）面板数据与横截面数据的比较

①面板数据有别于混合横截面数据的关键特征是，同一横截面数据的数据单位都被跟踪了一段特定的时期。

②由于面板数据要求同一单位不同时期的重复观测，所以要得到面板数据（特别是那些个人、家庭和企业的数据），比得到混合横截面数据更加困难。

③对同一观测单位观测一段时间，应该比横截面数据甚至混合横截面数据更有优越性。对同一单位的多次观测，能控制个人、企业等观测单位的某些观测不到的特征。

④面板数据的第二个优点是，它通常能够研究决策行为或结果中滞后的重要性。由于预期许多经济政策在一段时间之后才产生影响，所以面板数据所反映的信息就更有意义。

4.

- Causality and the notion of ceteris paribus

Definition of causal effect of x on y:

> "How does variable y change if variable x is changed but all other relevant factors are held constant"

- Most economic questions are ceteris paribus questions

其他条件不变

Interpretations of our regression result:

(from tutorial slides week 7)

1. Also remember we can only draw correlations rather than causation effect from observational data
2. Units of Measurement and Functional Form (with log or without log)

Model	Dep. Var.	Indep. Var.	β_1
Level-Level	y	x	$\Delta y = \beta_1 \Delta x$
Level-Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-Level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$

$\%\Delta x \rightarrow$ percentage change in x.

if you can't remember them during the test → take derivatives.

eg: log-level: $\frac{dy}{dx} = \frac{dy}{d\log x} \cdot \frac{d\log x}{dx} = \beta_1 \cdot \frac{1}{x} \rightarrow \Delta y = \beta_1 \frac{\Delta x}{x} = \frac{\beta_1}{100} \%\Delta x$

The difference between R^2 in SLR and R_j^2 in MLR.

SLR: $R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$

The interpretation of R-squared: how much of the variance in y can be explained by x.

MLR: $Var(\hat{\beta}_j) = \frac{\sigma^2}{SSR_j}$. SSR_j is hard to calculate directly, so we apply $SSR_j = SST_j(1-R_j^2)$ 用 R_j 是为了算 $Var(\hat{\beta}_j)$. R_j 本身没多大意义

Hence $Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$

The interpretation of R_j^2 : how much of the variance in x_j can be explained by $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$.

A brief summary of relevant distributions. (pay attention to degree of freedom).

SLR: Under SLR 1-6.

$\hat{\beta}_j \sim Normal(\beta_j, Var(\hat{\beta}_j))$, $\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim Normal(0,1)$

MLR: $\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$ $\left(se(\hat{\beta}_j) = \sqrt{Var(\hat{\beta}_j)} = \frac{\sigma^2}{SST_j(1-R_j^2)},\ \sigma^2 = Var(u_i) = \frac{\sum u_i^2}{n-k-1}\right)$

$\frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} \sim F_{(q,\,n-k-1)}$ (Under MLR 1-5)

Chow-test: (special F test) $\frac{[SSR_p - (SSR_1 + SSR_2)]/(k+1)}{(SSR_1 + SSR_2)/[n-2(k+1)]} \sim F_{(k+1,\,n-2(k+1))}$

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Proofs and Derivations to Know

1. Derivations of formulas for the OLS coefficient estimators in the SLR Model (In Week 2 Lecture)

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ ($\hat{y}_i$ is the predicted value of y_i)

residual $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$

choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize $\sum_{i=1}^{n} \hat{u}_i^2$

$$\min \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

By first-order condition:

1) w.r.t. $\hat{\beta}_0$: $n^{-1}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$ ①

着先得到 $\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$

展开后 $\sum y_i = n\bar{y}$, $\sum \hat{\beta}_0 = n\hat{\beta}_0$, $\sum \hat{\beta}_1 x_i = n\hat{\beta}_1 \bar{x}$

所以一开始同时乘以 n^{-1}.

2) w.r.t. $\hat{\beta}_1$: $n^{-1}\sum_{i=1}^{n} x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$ ②

From ①. $\frac{1}{n}\sum y_i - \frac{1}{n}\sum \hat{\beta}_0 - \frac{1}{n}\sum \hat{\beta}_1 x_i = 0$

$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0$ we get $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ ③

Plug ③ into ②: $\frac{1}{n}\sum x_i[y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i] = 0$

$\frac{1}{n}\sum x_i(y_i - \bar{y}) - \frac{1}{n}\hat{\beta}_1 \sum x_i(x_i - \bar{x}) = 0$ ④

$\because \sum x_i(y_i - \bar{y}) = \sum(x_i - \bar{x})(y_i - \bar{y}) + \bar{x}\sum(y_i - \bar{y})$

$\because \sum(y_i - \bar{y}) = n\bar{y} - n\bar{y} = 0$

$\therefore \sum x_i(y_i - \bar{y}) = \sum(x_i - \bar{x})(y_i - \bar{y})$ also $\sum(x_i - \bar{x})(y_i - \bar{y}) = \sum(x_i - \bar{x})y_i$

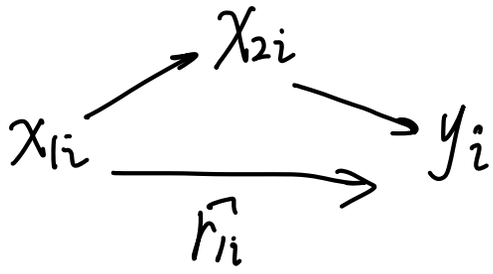
$\because \sum x_i(x_i - \bar{x}) = \sum(x_i - \bar{x})(x_i - \bar{x}) = \sum(x_i - \bar{x})^2$

$\therefore$ ④ becomes $\frac{1}{n}\sum(x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n}\hat{\beta}_1 \sum(x_i - \bar{x})^2$

If $\sum(x_i - \bar{x})^2 \neq 0$,

$$\begin{cases} \hat{\beta}_1 = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \dfrac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$

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2. Partialling Out Interpretation of MLR Model (In Week 4 Lecture)

Partial effect or ceteris paribus interpretations:

consider $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ → we have n of these, specifically, we can write $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}$

$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1 + \hat{\beta}_2 \Delta x_2$

$\therefore \Delta\hat{y} = \hat{\beta}_1 \Delta x_1$ if $\Delta x_2 = 0$. $\Delta\hat{y} = \hat{\beta}_2 \Delta x_2$ if $\Delta x_1 = 0$.

If we look at $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}$, partial out effect of x_{i2}

1) Do a sample regression of x_{i1} on x_{i2} to obtain residual $\hat{r}_{i1}$, so the effect of x_{i2} is partialled out

$\hat{r}_{i1}$ is a part of x_{i1} that is uncorrelated with x_{i2}

whether it is $\hat{r}_{i1}$ or $\hat{r}_{ii}$, please follow your instructor.

2) Regress y_i on $\hat{r}_{i1}$ to obtain $\hat{\beta}_1$

become SLR between y and x_1 after x_2 is partialled out.

$$\hat{\beta}_1 = \frac{\sum(\hat{r}_{i1} - \bar{\hat{r}}_{i1})y_i}{\sum(\hat{r}_{i1} - \bar{\hat{r}}_{i1})^2} = \frac{\sum \hat{r}_{i1} y_i}{\sum \hat{r}_{i1}^2}$$

→ $\hat{r}_{i1}$ 是 reg x_{i1} on x_{i2} 的 average residual.

we know $\sum_{i=1}^{n}(\hat{u}_i) = 0$, $\bar{\hat{r}}_{i1} = 0$

3. Derivation of OLS decomposition equation from OLS regression function, i.e., SST = SSE + SSR (In Week 2 Lecture)

$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$ total variation in dependent variables

$SSE = \sum(\hat{y}_i - \bar{y})^2$ variation explained by regression

$SSR = \sum \hat{u}_i^2$ variation not explained

$\hat{u}_i$

$$SST = \sum[(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})]^2$$

$$= \sum[(y_i - \hat{y}_i)^2 + (\hat{y}_i - \bar{y})^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y})]$$

$$= \sum \hat{u}_i^2 + \sum(\hat{y}_i - \bar{y})^2 + 2\sum \hat{u}_i(\hat{y}_i - \bar{y})$$

$$= SSR + SSE$$

$\because \sum(\hat{y}_i - \bar{y}) = 0$

$\therefore 2\sum \hat{u}_i(\hat{y}_i - \bar{y}) = 0$

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4. Proof of unibiasedness of OLS estimators (SLR and MLR)
(SLR: In Week 3 Lecture; MLR: In Week 4 Lecture)

SST
SSTx

SLR: step 1. $\tilde{\beta}_1 = \frac{\sum (x_i-\bar{x})y_i}{\sum (x_i-\bar{x})^2}$ define $SST_x = \sum_{i=1}^{n}(x_i-\bar{x})^2$ not SST $\left(\sum_{i=1}^{n}(y_i-\bar{y})^2\right)$

$\hat{\beta}_1 = \frac{\sum (x_i-\bar{x})y_i}{SST_x}$ SST_x is just some positive number

$\hat{\beta}_1$ existance follows SLR.3

step2. Replace y_i with $y_i = \beta_0 + \beta_1 x_i + u_i$.

$$\sum (x_i-\bar{x})y_i = \sum (x_i-\bar{x})(\beta_0+\beta_1 x_i+u_i)$$
$$= \beta_0 \sum (x_i-\bar{x}) + \beta_1 \sum x_i(x_i-\bar{x}) + \sum (x_i-\bar{x})u_i$$
$$= 0 + \beta_1 \sum (x_i-\bar{x})^2 + \sum (x_i-\bar{x})u_i = \beta_1 SST_x + \sum (x_i-\bar{x})u_i$$

$$\hat{\beta}_1 = \frac{\beta_1 SST_x + \sum (x_i-\bar{x})u_i}{SST_x} = \beta_1 + \frac{\sum (x_i-\bar{x})u_i}{SST_x}$$

define $w_i = \frac{(x_i-\bar{x})}{SST_x}$ fixed value (composition of xs, xs are known)

$\hat{\beta}_1 = \beta_1 + \sum w_i u_i$

step 3. $E(\hat{\beta}_1|x) = E(\beta_1 + \sum w_i u_i \mid x) = E(\beta_1|x) + \sum E(w_i u_i|x)$

$= E(\beta_1|x) + \sum w_i E(u_i|x) = E(\beta_1|x) = \beta_1$

conditioning on x, means that xs are fixed. → $E(u_i|x)=0$ SLR 4.

True for all xs. True without xs.

$\therefore E(\hat{\beta}_1) = \beta_1$

MLR: $\tilde{\beta}_j = \frac{\sum \hat{r}_{ij} y_i}{\sum \hat{r}_{ij}^2} = \frac{\sum \hat{r}_{ij}(\beta_0+\beta_1 x_{i1}+\cdots+\beta_j x_{ij}+\cdots+\beta_k x_{ik}+u_i)}{\sum \hat{r}_{ij}^2}$

$\because \sum \hat{r}_{ij} = 0$. $\sum \hat{r}_{ij} x_{ij} = \sum \hat{r}_{ij}(\hat{x}_{ij}+\hat{r}_{ij}) = \sum \hat{r}_{ij}\hat{x}_{ij} + \sum \hat{r}_{ij}^2 = \sum \hat{r}_{ij}^2$

$\sum \hat{r}_{ij} x_{il} = 0 \ (l \neq j)$

real dependent variable = estimated + residual

independent variables

$$\therefore \tilde{\beta}_j = \frac{\sum \hat{r}_{ij}\beta_j x_{ij} + \sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2} = \frac{\beta_j \sum \hat{r}_{ij} x_{ij} + \sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2} = \frac{\beta_j \sum \hat{r}_{ij}^2 + \sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2} = \beta_j + \frac{\sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2}$$

conditioning on x: $E(\hat{\beta}_j) = \beta_j + E\left[\frac{\sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2}\right] = \beta_j + \frac{\sum \hat{r}_{ij}}{\sum \hat{r}_{ij}^2} E(u_i)$

functions of xs. xs are fixed.

$\because E(u_i) = 0$

$\therefore E(\hat{\beta}_j) = \beta_j$

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5. Derivation of the variance of OLS slope estimators (SLR and MLR)
(SLR: In Week 3 Lecture; MLR: In Week 5 Lecture)

SLR: Assumption SLR.5 (Homoskedasticity): $Var(u_i|x_i) = Var(u) = \sigma^2$

$E(u^2|x) = E(u^2) = \sigma^2 \rightarrow Var(u|x) = E(u^2|x) - [E(u|x)]^2 = E(u^2|x)$

$$Var(y|x) = Var(\beta_0 + \beta_1 x + u|x) = Var(\beta_0|x) + Var(\beta_1 x|x) + Var(u|x)$$
$$= 0 + 0 + \sigma^2 = \sigma^2$$

$$E(y|x) = E(\beta_0 + \beta_1 x + u|x) = E(\beta_0|x) + E(\beta_1 x|x) + E(u|x)$$
$$= \beta_0 + \beta_1 x + 0 = \beta_0 + \beta_1 x$$

$$Var(\hat{\beta}_1|x) = Var(\beta_1 + \sum w_i u_i|x) = Var(\sum w_i u_i|x) = \sum w_i^2 Var(u_i|x)$$
$$= \sum w_i^2 \sigma^2$$

$\because \sum w_i^2 = \sum \frac{(x_i - \bar{x})^2}{(SST_x)^2} = \frac{\sum (x_i - \bar{x})^2}{(SST_x)^2} = \frac{SST_x}{(SST_x)^2} = \frac{1}{SST_x}$

$\therefore Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_x}$

MLR: Assumption MLR.5 (Homoskedasticity): $Var(u_i|x_i) = \sigma^2$

By MLR.1 and MLR.4: $E(y_i|x_i) = E(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u_i|x_i)$
$= \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$

Add MLR.5: $Var(y|x_1 \cdots x_k) = Var(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u_i|x_i)$
$= \sigma^2$

$\because \hat{\beta}_j = \beta_j + \frac{\sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2}$

$\therefore$ conditioning on x:

$$Var(\hat{\beta}_j) = Var\left(\frac{\sum \hat{r}_{ij} u_i}{\sum \hat{r}_{ij}^2}\right) = \left(\frac{1}{\sum \hat{r}_{ij}^2}\right)^2 Var(\sum \hat{r}_{ij} u_i) = \left(\frac{1}{\sum \hat{r}_{ij}^2}\right)^2 \left(\sum \hat{r}_{ij}^2\right) \sigma^2 = \frac{\sigma^2}{\sum \hat{r}_{ij}^2} = \frac{\sigma^2}{SSR_j}$$

$\because R_j^2 = 1 - \frac{SSR_j}{SST_j}$

$\therefore SSR_j = SST_j (1 - R_j^2)$

$\therefore Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j (1 - R_j^2)}$

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Important Things to know

1. Gauss-Markov Assumptions and Classical Linear Model assumptions (Gauss-Markov In Week 5 Lecture; CLM In Week 6 Lecture)

Gauss Markov assumptions in MLR:

- MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
- MLR.2: random sampling from the population
- MLR.3: no perfect collinearity in the sample → 否则会求解 → 解决方法: drop one of independent variables.
- MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
- MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)

both stronger and weaker than SLR4.

number of x_i increases → less remains in u.

- **Classical linear model (CLM) assumptions** = Gauss + Normality

$\underbrace{MLR.1 - MLR.5}$ "Gauss-Markov assumptions"

$\underbrace{MLR.1 - MLR.6}$ "Classical linear model (CLM) assumptions"

- u_i are independent and identically distributed (iid) as $Normal(0, \sigma^2)$

$$\hat{\beta}_j = \beta_j + \sum_{i=1}^{n} w_{ij} u_i \sim Normal[\beta_j, Var(\hat{\beta}_j)]$$

where linear combination of u_i, also normally distributed

The estimators are normally distributed around the true parameters with the variance that was derived earlier

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

2. Definition and meaning of the following statistical properties of estimators: (In Week 5 Lecture)

1) unibiasedness

$E(\hat{\beta}_j) = \beta_j$

2) minimum variance

Among all unbiased estimator, OLS gives the smallest variance. So, it is the 'best'. The proof is not required, but you can find it in the textbook appendix 3.

We show that, for any other linear unbiased estimator $\tilde{\beta}_1$ of β_1, $\text{Var}(\tilde{\beta}_1) \geq \text{Var}(\hat{\beta}_1)$, where $\hat{\beta}_1$ is the OLS estimator. The focus on $j = 1$ is without loss of generality.

For $\tilde{\beta}_1$ as in equation (3.60), we can plug in for y_i to obtain

$$\tilde{\beta}_1 = \beta_0 \sum_{i=1}^{n} w_{i1} + \beta_1 \sum_{i=1}^{n} w_{i1}x_{i1} + \beta_2 \sum_{i=1}^{n} w_{i1}x_{i2} + \ldots + \beta_k \sum_{i=1}^{n} w_{i1}x_{ik} + \sum_{i=1}^{n} w_{i1}u_i.$$

Now, since the w_{i1} are functions of the x_{ij},

$$\begin{aligned} \text{E}(\tilde{\beta}_1|\mathbf{X}) &= \beta_0 \sum_{i=1}^{n} w_{i1} + \beta_1 \sum_{i=1}^{n} w_{i1}x_{i1} + \beta_2 \sum_{i=1}^{n} w_{i1}x_{i2} + \ldots + \beta_k \sum_{i=1}^{n} w_{i1}x_{ik} + \sum_{i=1}^{n} w_{i1}\text{E}(u_i|\mathbf{X}) \\ &= \beta_0 \sum_{i=1}^{n} w_{i1} + \beta_1 \sum_{i=1}^{n} w_{i1}x_{i1} + \beta_2 \sum_{i=1}^{n} w_{i1}x_{i2} + \ldots + \beta_k \sum_{i=1}^{n} w_{i1}x_{ik} \end{aligned}$$

because $\text{E}(u_i|\mathbf{X}) = 0$, for all $i = 1, \ldots, n$ under MLR.2 and MLR.4. Therefore, for $\text{E}(\tilde{\beta}_1|\mathbf{X})$ to equal β_1 for any values of the parameters, we must have

$$\sum_{i=1}^{n} w_{i1} = 0, \sum_{i=1}^{n} w_{i1}x_{i1} = 1, \sum_{i=1}^{n} w_{i1}x_{ij} = 0, \quad j = 2, \ldots, k. \qquad [3.69]$$

Now, let $\hat{r}_{i1}$ be the residuals from the regression of x_{i1} on $x_{i2}, \ldots, x_{ik}$. Then, from (3.69), it follows that

$$\sum_{i=1}^{n} w_{i1}r_{i1} = 1 \qquad [3.70]$$

because $x_{i1} = \hat{x}_{i1} + \hat{r}_{i1}$ and $\sum_{i=1}^{n} w_{i1}\hat{x}_{i1} = 0$. Now, consider the difference between $\text{Var}(\tilde{\beta}_1|\mathbf{X})$ and $\text{Var}(\hat{\beta}_1|\mathbf{X})$ under MLR.1 through MLR.5:

$$\sigma^2 \sum_{i=1}^{n} w_{i1}^2 - \sigma^2 \Big/ \left(\sum_{i=1}^{n} \hat{r}_{i1}^2 \right). \qquad [3.71]$$

Because of (3.70), we can write the difference in (3.71), without σ^2, as

$$\sum_{i=1}^{n} w_{i1}^2 - \left(\sum_{i=1}^{n} w_{i1}\hat{r}_{i1} \right)^2 \Big/ \left(\sum_{i=1}^{n} \hat{r}_{i1}^2 \right). \qquad [3.72]$$

But (3.72) is simply

$$\sum_{i=1}^{n} (w_{i1} - \hat{\gamma}_1\hat{r}_{i1})^2, \qquad [3.73]$$

where $\hat{\gamma}_1 = \left(\sum_{i=1}^{n} w_{i1}\hat{r}_{i1} \right) \Big/ \left(\sum_{i=1}^{n} \hat{r}_{i1}^2 \right)$, as can be seen by squaring each term in (3.73), summing, and then canceling terms. Because (3.73) is just the sum of squared residuals from the simple regression of w_{i1} on $\hat{r}_{i1}$—remember that the sample average of $\hat{r}_{i1}$ is zero—(3.73) must be nonnegative. This completes the proof.

3) efficiency

OLS estimators are the best linear unbiased estimators (BLUEs). Since they give the smallest variances among all unbiased estimators, they are called efficient.

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3. Properties of the OLS sample regression equation
(SLR in Week 2 Lecture; MLR in Week 4 Lecture)

SLR:

$\sum_{i=1}^{n} \hat{u}_i = 0$ — Deviations from regression line sum up to zero

$\sum_{i=1}^{n} x_i \hat{u}_i = 0$ — Correlation between deviations and regressors is zero

$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ — Sample averages of y and x lie on regression line

MLR:

$\sum_{i=1}^{n} \hat{u}_i = 0$ — Deviations from regression line sum up to zero

$\sum_{i=1}^{n} x_{ij} \hat{u}_i = 0$ — Correlations between deviations and regressors are zero

$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \ldots + \hat{\beta}_k \bar{x}_k$ — Sample averages of y and of the regressors lie on regression line

4. Statistical properties of the OLS coefficient estimators $\hat{\beta}_j$, i.e. properties of $E(\hat{\beta}_j)$ and $Var(\hat{\beta}_j)$.
(In Week 4 and 5 lectures)

Expected:

1) unbiasedness
2) Overspecify the model by including irrelevant variables

eg: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$

But $\beta_3 = 0$ in the population

∴ No problem because $E(\hat{\beta}_3) = \beta_3 = 0$.

But may increase the variance of OLS estimators x_3 may have effect on other x_j, increase R_j thus increase $var(\hat{\beta}_j)$

3) Underspecify the model by omitting relevant variables

eg: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$

$\Rightarrow \tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

$\tilde{\beta}_1 = \frac{\sum (x_{i1} - \bar{x}_1) y_i}{\sum (x_{i1} - \bar{x}_1)^2} = \frac{\sum (x_{i1} - \bar{x}_1)(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i)}{\sum (x_{i1} - \bar{x}_1)^2}$ 这里只有一个 independent variable，写 $\frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2}$ 一样.

$= \frac{\beta_0 \sum (x_{i1} - \bar{x}_1)}{\sum (x_{i1} - \bar{x}_1)^2} + \frac{\beta_1 \sum x_{i1}(x_{i1} - \bar{x}_1)}{\sum (x_{i1} - \bar{x}_1)^2} + \frac{\beta_2 \sum (x_{i1} - \bar{x}_1) x_{i2}}{\sum (x_{i1} - \bar{x}_1)^2} + \frac{\sum (x_{i1} - \bar{x}_1) u_i}{\sum (x_{i1} - \bar{x}_1)^2}$

$= 0 + \beta_1 + \beta_2 \frac{\sum \cdots}{\sum \cdots} + \frac{\sum \cdots u_i}{\sum \cdots}$

$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \frac{\sum (x_{i1} - \bar{x}_1) x_{i2}}{\sum (x_{i1} - \bar{x}_1)^2}$ conditioning on x_i fixed

$= \beta_1 + \beta_2 \delta_1$

$E(\tilde{\beta}_1) > \beta_2$ = upward bias

$E(\tilde{\beta}_1) < \beta_2$ = downward bias

Omitted Variable Bias.

Variance: efficiency

0. R^2

take SLR as an example:

$SST=\sum(y_i-\bar{y})^2$ $SSE=\sum(\hat{y}_i-\bar{y})^2$ $SSR=\sum_{i=1}^{n}\hat{u}_i^2$

total sum of squares explained sum of squares residual sum of squares

$SST=SSE+SSR$

prove:

$$SST=\sum_{i=1}^{n}(y_i-\bar{y})^2=\sum_{i=1}^{n}[(y_i-\hat{y}_i)+(\hat{y}_i-\bar{y})]^2$$

$$=\sum_{i=1}^{n}[\hat{u}_i+(\hat{y}_i-\bar{y})]^2$$

$$=\sum_{i=1}^{n}\hat{u}_i^2+2\sum\hat{u}_i(y_i-\bar{y})+\sum(y_i-\bar{y})^2$$

$\sum\hat{u}_i\hat{y}_i=0$ $\sum\hat{u}_i\bar{y}=0$

thus $=\sum_{i=1}^{n}\hat{u}_i^2+\sum(y_i-\bar{y})^2=SSE+SSR$

$R^2=\frac{SSE}{SST}=1-\frac{SSR}{SST}$ $0\le R^2\le 1$

↳ the share of explained sum of squares in terms of total

⇒ the higher, the better → statistical significance

$R^2\uparrow$, model is more accurate (Attention: R^2 only shows whether the model fits data. It can't show causality) → to show causality → Ceteris paribus

still, a small R^2 doesn't mean the model is bad, the key/most important thing for one model is zero conditional mean

1. β_j

Statistical Inference (P105-148 on the textbook)

① About $\beta_j/\hat{\beta}_j$

$\hat{\beta}_j$ of course is not β_j, it's an estimated value based on data.

How should we approach β_j? First, understand $\hat{\beta}_j$'s distribution

$\hat{\beta}_j=\beta_j+\sum_{i=1}^{n}w_{ij}u_i$ And we know: MLR.1 → MLR.4: $E(\hat{\beta}_j)=\beta_j$

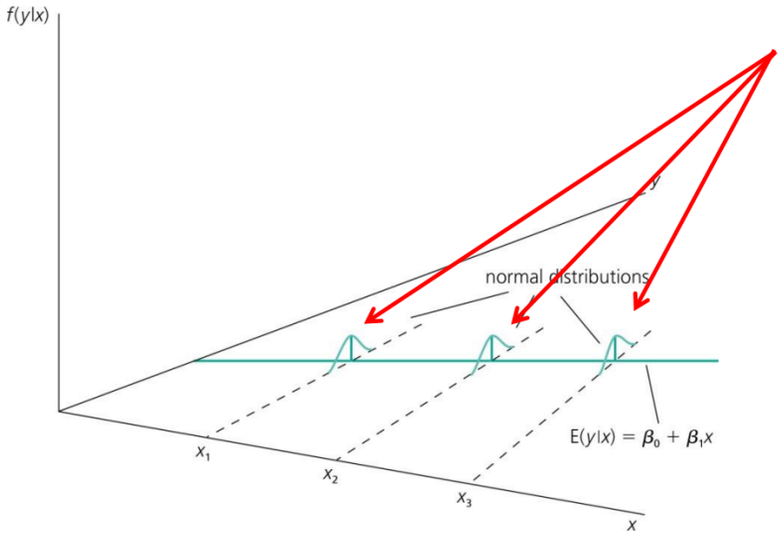
MLR.1 → MLR.5: $Var(\hat{\beta}_j)=\frac{\sigma^2}{SST_j(1-R_j^2)}$

But, given all these equations, still can't know $\hat{\beta}_j$'s distribution

⇒ MLR.6 (normality of error terms)

$u_i\sim N(0,\sigma^2)$ This strong assumption pins down $\hat{\beta}_j$'s distribution ⇒ normal dis

$u_i \sim N(0, \sigma^2)$ independently of $x_{i1}, x_{i2}, \ldots, x_{ik}$



It is assumed that the unobserved factors are normally distributed around the population regression function.

The form and the variance of the distribution does not depend on any of the explanatory variables.

It follows that:

$y|\mathbf{x} \sim N(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k, \sigma^2)$

- **Discussion of the normality assumption**
  - The error term is the sum of "many" different unobserved factors. Sums of independent factors are normally distributed (CLT)
  - The normality of the error term is an empirical question
  - Examples where normality cannot hold:
    - Wages (nonnegative; minimum wage)
    - Unemployment (indicator variable, takes on only 1 or 0)
  - In some cases, normality can be achieved through transformations of the dependent variable (e.g. use log(wage) instead of wage)
  - At least the error distribution should be "close" to normal. In many cases, normality is questionable or impossible by definition

2. Distribution of β_j

MLR.1 – MLR.5 Gauss-Markov assumptions

MLR.1 – MLR.6 CLM assumptions (classical linear model)

Under CLM:

$\hat{\beta}_j \sim N[\beta_j, Var(\hat{\beta}_j)]$ $\quad Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$

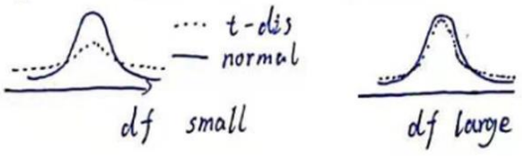
thus, $\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim N(0,1)$ the same with previous situation, $sd(\hat{\beta}_j)$ requires σ, which is unknown

$\Rightarrow$ use $\hat{\sigma}$ to estimate $\Rightarrow$ change from normal to t-distribution

$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$ In this case, degree of freedom = $n-k-1$

= n observations – K explanatory variable – 1 (number of β_0, which we only have one)

if $df > 120$, t-dis is very similar to a normal dis



df small df large

- **Null hypothesis** the equal sign "=" should always be in H_0.

$$H_0: \beta_j = 0$$

The population parameter is equal to zero, i.e. after controlling for the other independent variables, there is no effect of x_j on y

- **t-statistic (or t-ratio)**

The t-statistic will be used to test the null hypothesis.

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

The farther the estimated coefficient is away from zero, the less likely it is that the null hypothesis holds true. But what does "far" away from zero mean?

The t-statistic measures how many estimated standard deviations the estimated coefficient is away from zero.

- **Distribution of the t-statistic <u>if the null hypothesis is true</u>**

$$t_{\hat{\beta}_j} = \hat{\beta}_j / se(\hat{\beta}_j) = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim t_{n-k-1}$$

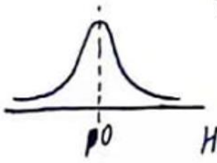
We calculate t-statistic based on the assumption that H_0 is true.

we can never support H_0. only reject or fail to reject H_0. (type II error).

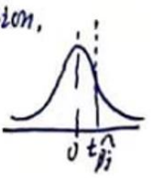
3. hypothesis testing

Use of hypothesis testing: too see/check whether we can reject the null hypothesis H_0. (null hypothesis): In this Chapter. $H_0 = \beta_j = 0$, that means, we want to check whether $\beta_j = 0$ or β_j do exist.

How?



By turning $\hat{\beta}_j$ into a t distribution,
if $\beta_j = 0$, $t_{\hat{\beta}_j}$ is likely to be in an interval around 0.

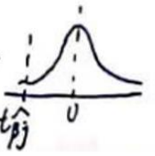


likely

However, if $t_{\hat{\beta}_j}$ is far from 0, this is a highly unlikely situation if $\beta_j = 0$. Thus, $\beta_j \neq 0$, we reject null hypothesis



or



unlikely

So, if given the criteria in H_0, the prob of arriving this specific t_{β_j} is small, reject H_0

But how 'small' is small?

We call the method of measuring small as ~~critical value~~ 'significance level'

the significance level is eg: 5%, not 95%.

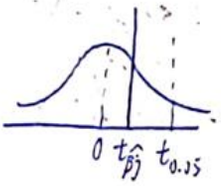
For example: $H_0: \beta_j = 0$ at 5% significance level
$H_A: \beta_j > 0$



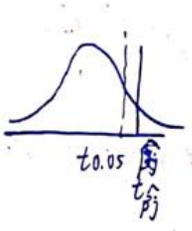
the shadow area is α
since at 5% significance level, the probability for $\hat{\beta}_j$ to fall in this range is 5%

Check t table and get corresponding $t_{0.05}$ for the border line

What does 5% significance level tells us?



if $\hat{\beta}_j$ falls in this area:
$\hat{\beta}_j$ is likely to be 0 since $\hat{\beta}_j$ is close to 0
Don't reject H_0



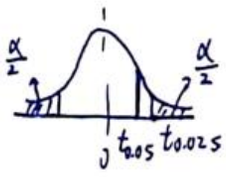
if falls in this area
$t_{\hat{\beta}_j} > t_{0.05}$
this means if $\beta_j = 0$, the Prob of having such $\hat{\beta}_j$ is smaller than 5%
we reject H_0

the same for $H_0: \beta_j = 0$ $H_1: \beta_j < 0$

How about two-tailed testing?

$H_0: \beta_j = 0$ $H_1: \beta_j \neq 0$

Attention, since graph change. but significance level doesn't change



now each shadow area represents 2.5% possibilities instead of 5%

we can see that

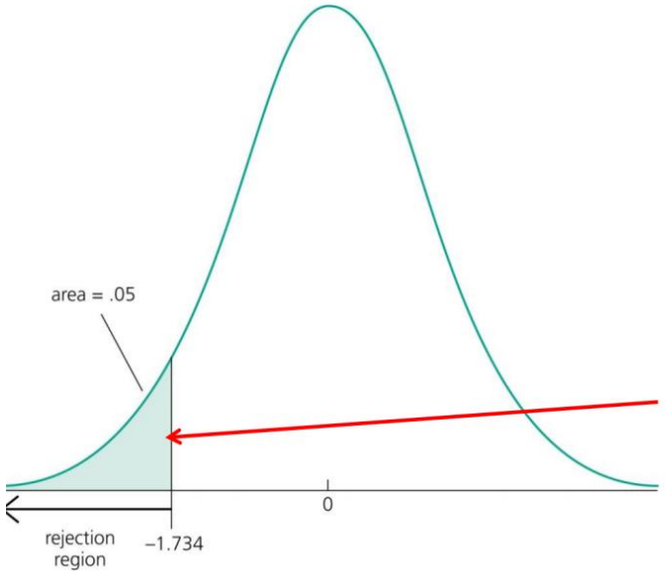
$t_{0.05} < t_{0.025}$

$t_{\hat{\beta}_j}$ should > $t_{0.025}$ to ~~move~~ reject H_0 (more difficult)

☆ $2t_{0.025} \neq t_{0.05}$.

- **Testing against one-sided alternatives (less than zero)**

  5% rejection rule for alternative $H_1 : \beta_j < 0$ with 18 *df.*



Reject the null hypothesis in favour of the alternative hypothesis if the estimated coefficient is "too small" (i.e. smaller than a critical value).

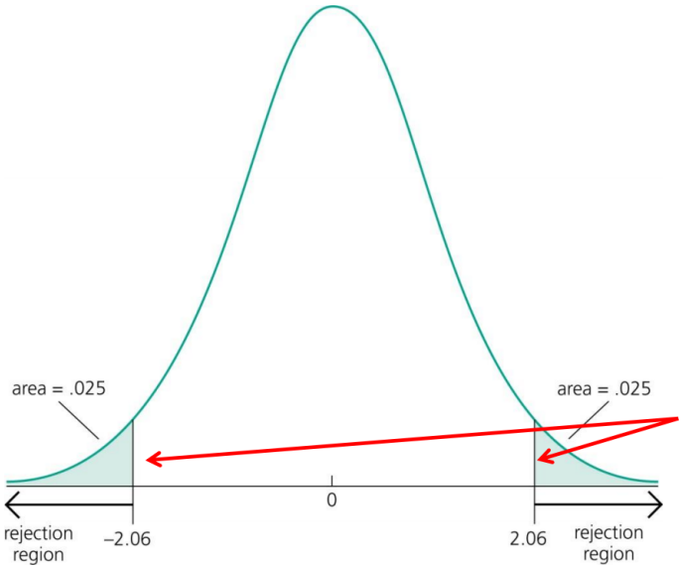
Construct the critical value so that, if the null hypothesis is true, it is rejected in, for example, 5% of the cases.

In the given example, this is the point of the t-distribution with 18 degrees of freedom so that 5% of the cases are below the point.

Reject if t-statistic less than -1.734

- **Testing Against Two-Sided Alternatives**

  5% rejection rule for alternative $H_1 : \beta_j \neq 0$ with 25 *df.*



Reject the null hypothesis in favour of the alternative hypothesis if the absolute value of the estimated coefficient is too large.

Construct the critical value so that, if the null hypothesis is true, it is rejected in, for example, 5% of the cases.

In the given example, these are the points of the t-distribution so that 5% of the cases lie in the two tails.

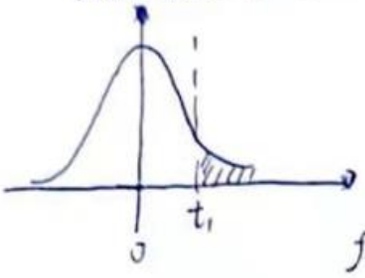
Reject if absolute value of t-statistic is less than -2.06 or greater than 2.06

4. /5. P-value approach/confidence interval

④ p-value approach

Just now we introduced how to conduct hypothesis testing using 'critical value'.

Now Lest's make a simple comparison:



significance level = 5%

critical value:

find $t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - 0}{se(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$

find t_1 according to t table

(df = n-K-1, 5%)

compare $t_{\hat{\beta}_j}$ and t_1

p-value:

$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$



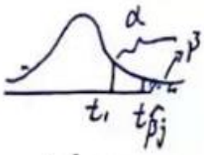
check t table

find out the probability

if falls in the shadow area

$\Rightarrow P(|T| > t_{\hat{\beta}_j})$

if $P(|T| > t_{\hat{\beta}_j}) < 0.05$:



$t_1 < t_{\hat{\beta}_j}$

reject H_0

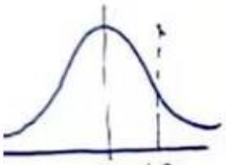
if $P(|T| > t_{\hat{\beta}_j}) > 0.05$



$t_1 > t_{\hat{\beta}_j}$

not reject H_0

⑤ confidence interval:



$t_{\hat{\beta}_j}$ could be anywhere,

we want to use $t_{\hat{\beta}_j}$ to estimate

β_j. how?

Remember $t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)}$

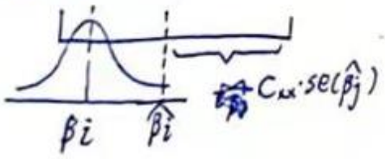
cofidence interval =

$(\hat{\beta}_j - c_{0.05} \cdot se(\hat{\beta}_j),\ \hat{\beta}_j + c_{0.05} \cdot se(\hat{\beta}_j))$

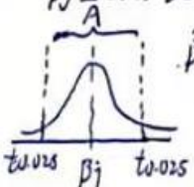
$\beta_i = \hat{\beta}_j - t_{\hat{\beta}_j} se(\hat{\beta}_j)$

we don't know $\beta_j \Rightarrow$ we don't know $t_{\hat{\beta}_j}$

but we can estimate



take $\hat{\beta}_j \pm c_{0.05} \cdot se(\hat{\beta}_j)$ as an example:



$\hat{\beta}_j$ is highly likely to show up in A range (95%)

if it is the case, the $|\beta_j - \hat{\beta}_j| = se(\hat{\beta}_j) \cdot c_{0.05}$

and for every $\hat{\beta}_j$ in this distribute, [illegible]

$(\hat{\beta}_j - c_{0.05} se(\hat{\beta}_i),\ \hat{\beta}_j + c_{0.05} \cdot se(\hat{\beta}_i))$ includes

- **Computing p-values for t-tests**
  - If the significance level is made smaller and smaller, there will be a point where the null hypothesis cannot be rejected anymore
  - The smallest significance level at which the null hypothesis is still rejected, is called the p-value of the hypothesis test
  - **p-value in two-sided test**,

$$p\text{-value} = P(|T| > |t|)$$

where t is the value of the $t - statistic$ and T is a random variable with the t_{n-k-1} distribution.

  - Perhaps its most useful interpretation is the following: the p-value is the probability of observing a $t - statistic$ as extreme as we did if the null hypothesis is true

Guidelines for discussing economic and statistical significance

- t testing is purely about statistical significance. It does not directly speak to the issue of whether a variable has a practically, or economically, large effect.
- **Statistical significance** depends on $t_{\hat{\beta}_j}$; **Economic (practically) significance** depends on the size (and sign) of $\hat{\beta}_j$
- If a variable is statistically significant, discuss the magnitude of the coefficient to get an idea of its economic or practical importance
- The fact that a coefficient is statistically significant does not necessarily mean it is economically or practically significant!
- If a variable is statistically and economically important but has the "wrong" sign, the regression model might be misspecified
- If a variable is statistically insignificant at the usual levels (10%, 5%, 1%), one may think of dropping it from the regression

6. linear combination of parameters

for example:

$\log(wage) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + u$

We try to find the relation between β_1, β_2

more specific, $H_0: \beta_1 - \beta_2 = 0$

$H_1: \beta_1 - \beta_2 < 0$

for this H_0, $t = \dfrac{\hat{\beta}_1 - \hat{\beta}_2}{se(\hat{\beta}_1 - \hat{\beta}_2)}$

However, $se(\hat{\beta}_1 - \hat{\beta}_2)$ can't be calculated directly

$se(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{Var(\hat{\beta}_1 - \hat{\beta}_2)} = \sqrt{\hat{var}(\hat{\beta}_1) + \hat{var}(\hat{\beta}_2) - 2cov(\hat{\beta}_1, \hat{\beta}_2)}$

- **Difficult to compute with standard regression output**

$$se(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{\widehat{Var}(\hat{\beta}_1 - \hat{\beta}_2)}$$

$$= \sqrt{\widehat{Var}(\hat{\beta}_1) + \widehat{Var}(\hat{\beta}_2) - 2\widehat{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$$

Usually not available in regression output, need to be estimated separately in Stata

```
. vce

Covariance matrix of coefficients of regress model
```

e(V)	jc	univ	exper	_cons
jc	.00004663			
univ	1.928e-06	5.330e-06		
exper	-1.718e-08	3.933e-08	2.480e-08	
_cons	-.00001741	-.00001573	-3.105e-06	.00044353

One can calculate $se(\hat{\beta}_1 - \hat{\beta}_2) = 0.0069357$

What if the equation too complex?

like: $H_0: \beta_2 = 3\beta_3$ $\quad$ $\mathrm{Var}(\hat{\beta}_2 - 3\hat{\beta}_3)$ is too complex

$H_1: \beta_2 - 3\beta_3 \neq 0$

another method:

$\theta_2 = \beta_2 - 3\beta_3 \Rightarrow \beta_2 = \theta_2 + 3\beta_3$

log wage $= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$

$= \beta_0 + \beta_1 X_1 + \theta X_2 + \beta_3 X_3 + u$

now only need to show θ alone

- **Alternative method** (Rewrite the model)

Define $\theta_1 = \beta_1 - \beta_2$ and test $H_0 : \theta_1 = 0$ against $H_1 : \theta_1 < 0$.

Insert into original regression

$$\log(wage) = \beta_0 + (\theta_1 + \beta_2)jc + \beta_2 univ + \beta_3 exper + u$$

$$= \beta_0 + \theta_1 jc + \beta_2 (jc + univ) + \beta_3 exper + u$$

a new regressor (= total years of college and university)

$$se(\hat{\theta}_1) = se(\hat{\beta}_1 - \hat{\beta}_2)$$

F test (multiple linear restrictions)

$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$

$H_0: \beta_3 = 0, \beta_4 = 0, \beta_5 = 0$

H_1: H_0 not true

use F test to check H_0

unrestricted model: the origin one

restricted model: $\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + u$ (exclude variables in H_0)

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)}$$

q: how many variables in H_0 (here is 3)

k: total explanatory variables

And since $SSR_r = (1-R_r^2)SST$

$SSR_{ur} = (1-R_{ur}^2)SST$

$R_r < R_{ur}$ (less explanatory variable)

thus $SSR_r > SSR_{ur}$

SST remain unchanged (since observation is the same)

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1-R_{ur}^2)/(n-k-1)}$$

- **Relationship between F and t statistics**

When we apply the F statistic to the case of testing significance of a single independent variable, i.e., $q = 1$

$$F_{1,n-k-1} = t_{n-k-1}^2$$

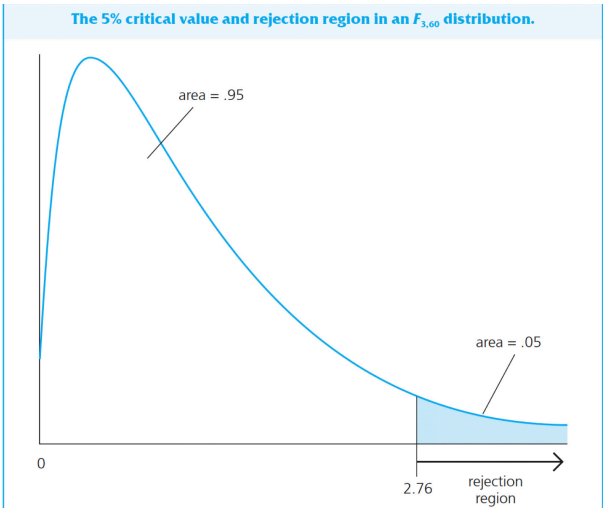
- **R-squared form of the F Statistic**

The same dependent variable is used in the two regression,

$$\begin{aligned} SSR_r &= (1 - R_r^2)SST \\ SSR_{ur} &= (1 - R_{ur}^2)SST \end{aligned}$$

therefore,

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)}$$



F-test is a one-sided test.

Binary variable = dummy variable dummy variable trap.

Week 8 Binary and Categorical Variables

For binary and categorical variables, always have an included group

$$\log(wage_i) = \beta_0 + \delta_0 Female_i + \delta_1 NotFemale_i + \beta_1 Educ_i + u_i$$

✗ $Female_i + NotFemale_i = 1$, perfect multicollinearity

1. Binary: $y_i = \beta_0 + \beta_1 X + \beta_2 Z + u$ ($X = 0$ or 1)

β_1 Interpretation: what is the average difference between observations where X = 1 and where X = 0.

$$\beta_1 = E[y|X=1, Z] - E[y|X=0, Z]$$

2. Categorical:

$$\log(wage_i) = \beta_0 + \delta_1 marrmale_i + \delta_2 singfem_i + \delta_3 marrfem_i + \beta_1 Educ_i + \beta_2 exper_i + \beta_3 tenure_i + u_i \quad (*)$$

(included group: $singmale_i$)

Intepreation :

if $marrmale_i = singfem_i = marrfem_i = 0$

Note that we still exclude one group : Single men

- interpretation of δ_1: average percent difference in wages between married and single men, holding educ, exper, tenure constant
- interpretation of δ_2: average percent difference in wages between single females and single men, holding educ, exper, tenure constant

δ_3 is similar

2.1. test H_0 such as $\delta_2 = \delta_3$

① Rewrite the model (change the included group)

② Treat the model as interactive (Also a kind of rewrite)

Rewrite model has been discussed in β_{21}

$$\log(wage_i) = \beta_0 + \delta_1 marrmale_i + \delta_2 female_i + \delta_3 marrfem_i + \beta_1 educ_i + \beta_2 exper_i + \beta_3 tenure_i + u_i \quad (12)$$

$$E[\log(wage_i) | female = 1, married = 1, educ, exper, tenure] - E[\log(wage_i) | female = 1, married = 0, educ, exper, tenure] = \delta_2 + \delta_3 - \delta_2 = \delta_3 \quad (13)$$

in function (13) : $\delta_3 - \delta_2$ → the mean percent difference in wages between married and single women, holding educ, exper, tenure constant.

now: keep female = 1, δ_3

3. Test for difference in regression functions across groups

$$\log(wage_i) = \beta_0 + \delta_1 female_i + \beta_2 Educ_i + \beta_3 exper_i + \beta_4 tenure_i + \delta_2 female_i * Educ_i + \delta_3 female_i * exper_i + \delta_4 female_i * tenure_i + u_i \quad (28)$$

test whether $\delta_1, \delta_2, \delta_3, \delta_4 = 0$. → F test : $H_0: \delta_1 = \delta_2 = \delta_3 = \delta_4 = 0$.

4. Chow-test.

Chow Test, which is a special F-test:

The ordinary F-test is too complicated if there are dummy variables, independent variables, and relative interaction terms.

Alternative way to compute F-statistic: Chow-Test

1) Pool the data and estimate a single regression. This is the restricted model, and produces the restricted SSR. Call this the pooled SSR, SSR_p
2) Estimate the regressions on the two groups (say, 1 and 2) separately. Get the SSR_1 and SSR_2. The unrestricted SSR is $SSR_1 + SSR_2$ (and this is the same as the regression that includes the full set of interactions).

$$\text{Chow-test}: \frac{[SSR_p - (SSR_1 + SSR_2)]/(k+1)}{(SSR_1 + SSR_2)/[n-2(k+1)]} \sim F_{(k+1,\, n-2(k+1))}$$